Data Flow Diagram

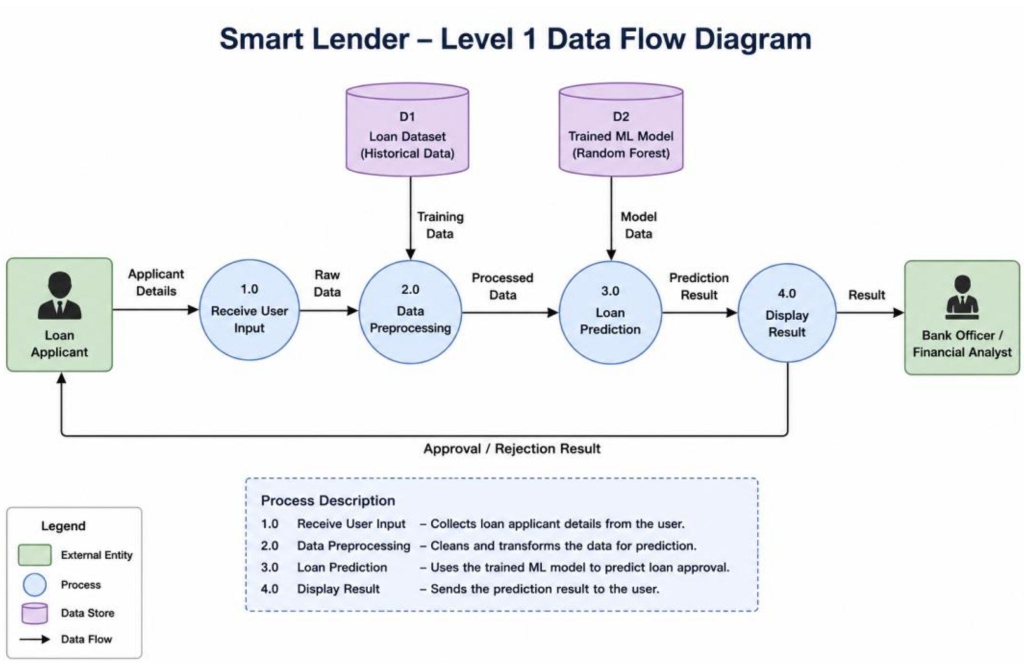
Date	06 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	2 Marks

Data Flow Diagram

The diagram below shows how data moves through the Smart Lender loan approval prediction system - between the applicant (external entity), the internal processes (application intake, preprocessing, and prediction), and the data stores that hold application data and the trained ML model.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person or system outside Smart Lender that sends or receives data (e.g., Applicant/User).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Preprocess Data, Run Prediction).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Application Datastore, Model Store).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.



Process Description

1. Application Intake - Collects the loan application form (Gender, Married, Dependents, Education, Self Employed, Income, Loan Amount, Loan Term, Credit History, Property Area) from the Applicant and validates it before storing it in the Application Datastore.

2. Preprocessing - Reads the stored applicant record, encodes categorical fields, and uses the Scaler Store (scale1.pkl) to scale the input features into a numeric feature vector.

3. Prediction Engine - Loads the trained Random Forest model from the Model Store (rdf.pkl), runs the prediction on the scaled feature array, and generates the loan approval/rejection decision that is returned to the Applicant.